```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <time.h>

#define BUFFER_SIZE 4096

// Function to copy using low-level system calls (open, read, >
void copy_low_level(const char *src, const char *dest) {
    int fd_src = open(src, O_RDONLY);
    if (fd_src < 0) {
        perror("Source open failed");
        exit(1);
    }

    int fd_dest = open(dest, O_WRONLY | O_CREAT | O_TRUNC, 064>
    if (fd_dest < 0) {
        perror("Destination open failed");
        close(fd_src);
        exit(1);
    }

    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    // Optional use of lseek (e.g., ensuring we start from beg>
    lseek(fd_src, 0, SEEK_SET);

    while ((bytes_read = read(fd_src, buffer, BUFFER_SIZE)) > >
        bytes_written = write(fd_dest, buffer, bytes_read);
        if (bytes_written != bytes_read) {
            perror("Write failed");
```

```c
// Function to copy using standard library functions (fopen, f>
void copy_standard(const char *src, const char *dest) {
    FILE *f_src = fopen(src, "rb");
    if (!f_src) {
        perror("Source fopen failed");
        exit(1);
    }

    FILE *f_dest = fopen(dest, "wb");
    if (!f_dest) {
        perror("Destination fopen failed");
        fclose(f_src);
        exit(1);
    }

    char buffer[BUFFER_SIZE];
    size_t bytes_read;

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, f_src))>
        fwrite(buffer, 1, bytes_read, f_dest);
    }

    fclose(f_src);
    fclose(f_dest);
}

int main() {
    const char *source_file = "source_test.txt";
    const char *dest_low = "dest_low.txt";
    const char *dest_std = "dest_std.txt";

    // Create a dummy source file with some data to test copyi>
```

```c
    // Create a dummy source file with some data to test copyi>
    FILE *fp = fopen(source_file, "w");
    if (fp) {
        for (int i = 0; i < 500000; i++) {
            fprintf(fp, "This is a test line for file copying >
        }
        fclose(fp);
    }

    clock_t start, end;
    double time_low, time_std;

    // 1. Benchmark Low-Level I/O
    start = clock();
    copy_low_level(source_file, dest_low);
    end = clock();
    time_low = ((double)(end - start)) / CLOCKS_PER_SEC;

    // 2. Benchmark Standard Library I/O
    start = clock();
    copy_standard(source_file, dest_std);
    end = clock();
    time_std = ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("=== File Copy Performance Comparison ===\n");
    printf("Low-Level I/O (open/read/write) Time : %.6f second>
    printf("Standard Library (fopen/fread/fwrite) Time: %.6f s>

    return 0;
}
```

```
PS C:\Users\Gireesh> wsl
gireesh@Kannigantiii:/mnt/c/Users/Gireesh$ cd ~/oslab
gireesh@Kannigantiii:~/oslab$ nano file_copy.c
gireesh@Kannigantiii:~/oslab$ gcc file_copy.c -o file_copy
./file_copy
=== File Copy Performance Comparison ===
Low-Level I/O (open/read/write) Time : 0.080256 seconds
Standard Library (fopen/fread/fwrite) Time: 0.088051 seconds
gireesh@Kannigantiii:~/oslab$
```